Class 12 · Mathematics

Chapter 1: Sets

Worksheet — Sets (NIOS Official)

Total marks: **30** · Questions: **10**

Instructions: This is the NIOS Senior Secondary Mathematics (311) official Worksheet-1 for Lesson 1: Sets. Attempt all questions. Show your reasoning and intermediate sets clearly.

Q1. Write three different sets in the roster method by taking different objects in your surroundings. [2 marks]

Q2. Develop any three sets in the set-builder form by using any type of numbers in the number system. [2 marks]

Q3. If $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$ and $B = \{5, 6, 7\}$, find (i) $A - B$ and (ii) $B - A$. Are they equal? [3 marks]

Q4. If $A = \{1, 3, 5, 7, 9\}$, $B = \{5, 6, 7, 8\}$ and $C = \{7, 8, 9\}$, find (i) $A \cap (B \cap C)$ and (ii) $A \cup (B \cup C)$. [3 marks]

Q5. Given $A = \{x : x \in Z \text{ and } x < 6\}$ and $B = \{y : y \text{ is a prime number}, y < 10\}$, find (i) $A \cup B$ and (ii) $A \cap B$. [3 marks]

Q6. Write down all the subsets of (i) A = {a, b} and (ii) B = {1, 2, 3}. Observe the relationship between the number of elements and the number of subsets. [3 marks]

Q7. Give an example of sets A and B where A is a subset of B. Draw Venn diagrams for A – B and B – A when $A \subseteq B$. [3 marks]

Q8. If U = {1, 2, 3, 4, 5, 6, 7, 8}, A = {1, 3, 5, 7} and B = {2, 4, 6, 8}, verify (i) $(A \cup B)' = A' \cap B'$ and (ii) $(A \cap B)' = A' \cup B'$. [4 marks]

Q9. Let N be the universal set of natural numbers and let A and B be its subsets given by $A = \{x : x \in N \text{ and } x < 10\}$ and $B = \{x : x \in N \text{ and } x \text{ is a multiple of } 5\}$. Find the complements of A and B. [3 marks]

Q10. Let U = {2, 4, 6, 8, 10, 12, 14, 16}, A = {2, 4, 6, 8}, B = {10, 12, 14, 16}. Verify (i) $(A')' = A$, (ii) $A \cup A' = U$, (iii) $(B')' = B$. [4 marks]

Answer Key

Worksheet — Sets (NIOS Official)

Q1. Write three different sets in the roster method by taking different objects in your surroundings.

Answer: Example answers: (1) Set of three primary colours: {red, yellow, blue}. (2) Set of vowels in the word 'mathematics': {a, e, i}. (3) Set of even numbers in the first 10 natural numbers: {2, 4, 6, 8, 10}. Any three distinct sets with their elements listed inside curly braces are acceptable.

Look for three sets with elements explicitly listed, comma-separated, inside braces.

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Q2. Develop any three sets in the set-builder form by using any type of numbers in the number system.

Answer: Example answers: (1) $\{x : x \in N$ and x is odd, $x < 10\} = \{1, 3, 5, 7, 9\}$. (2) $\{x : x \in Z$ and $-3 \leq x \leq 3\} = \{-3, -2, -1, 0, 1, 2, 3\}$. (3) $\{x : x \in Q$ and $0 < x < 1$, denominator $2\} = \{1/2\}$. Any three sets stated in set-builder form with a clear property are acceptable.

Set-builder form: $\{x : P(x)\}$ where P is the defining property. Include the universe (N, Z, Q, R) and the constraint.

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Q3. If $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$ and $B = \{5, 6, 7\}$, find (i) $A – B$ and (ii) $B – A$. Are they equal?

Answer: (i) $A - B = \{1, 2, 3, 4, 8\}$ (elements in A but not in B). (ii) $B - A = \varnothing$ (every element of B is also in A). $A - B \neq B - A$ in general. Here $A - B$ has 5 elements while $B - A$ has 0, so they are clearly not equal.

Set difference is not commutative: $A – B$ picks elements of A absent from B; $B – A$ picks elements of B absent from A.

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Q4. If $A = \{1, 3, 5, 7, 9\}$, $B = \{5, 6, 7, 8\}$ and $C = \{7, 8, 9\}$, find (i) $A \cap (B \cap C)$ and (ii) $A \cup (B \cup C)$.

Answer: (i) $B \cap C = \{7, 8\}$. $A \cap (B \cap C) = \{1, 3, 5, 7, 9\} \cap \{7, 8\} = \{7\}$. (ii) $B \cup C = \{5, 6, 7, 8, 9\}$. $A \cup (B \cup C) = \{1, 3, 5, 7, 9\} \cup \{5, 6, 7, 8, 9\} = \{1, 3, 5, 6, 7, 8, 9\}$.

Apply intersection / union one step at a time; $\cup$ and $\cap$ are both associative and commutative.

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Q5. Given $A = \{x : x \in Z$ and $x < 6\}$ and $B = \{y : y$ is a prime number, $y < 10\}$, find (i) $A \cup B$ and (ii) $A \cap B$.

Answer: $A = \{..., -2, -1, 0, 1, 2, 3, 4, 5\}$ (all integers strictly less than 6). $B = \{2, 3, 5, 7\}$. (i) $A \cup B = \{..., -2, -1, 0, 1, 2, 3, 4, 5, 7\}$. (ii) $A \cap B = \{2, 3, 5\}$ (the primes less than 6 that are in A).

Note that $7 \in B$ but $7 \notin A$ (since $7 \geq 6$); and 2, 3, 5 are in both A and B.

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Q6. Write down all the subsets of (i) A = {a, b} and (ii) B = {1, 2, 3}. Observe the relationship between the number of elements and the number of subsets.

Answer: (i) Subsets of A: ∅, {a}, {b}, {a, b}. That is 4 = 2^2 subsets. (ii) Subsets of B: ∅, {1}, {2}, {3}, {1, 2}, {1, 3}, {2, 3}, {1, 2, 3}. That is 8 = 2^3 subsets. Observation: a set with n elements has exactly 2^n subsets — including the empty set and the set itself. This is the size of the power set: |P(A)| = 2^|A|.

Each element can either belong to a subset or not — giving 2 choices per element. n elements ⇒ 2^n possible subsets.

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Q7. Give an example of sets A and B where A is a subset of B. Draw Venn diagrams for A – B and B – A when A ⊆ B.

Answer: Example: A = {1, 2} and B = {1, 2, 3, 4}. Then A ⊆ B. A − B = ∅ (every element of A is also in B, so there's nothing to remove). B − A = {3, 4} (the elements of B that are not in A). Venn diagram: draw two circles, one (B) larger and one (A) entirely INSIDE B. The region of A unshaded is A − B = ∅. The region of B outside A is B − A. The picture shows that when A ⊆ B, the difference A − B vanishes while B − A is the 'shell' of B outside A.

When A ⊆ B, A is entirely inside B in the Venn diagram. So A – B = ∅ and B – A is the part of B outside A.

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Q8. If U = {1, 2, 3, 4, 5, 6, 7, 8}, A = {1, 3, 5, 7} and B = {2, 4, 6, 8}, verify (i) (A ∪ B)' = A' ∩ B' and (ii) (A ∩ B)' = A' ∪ B'.

Answer: A' = U − A = {2, 4, 6, 8}. B' = U − B = {1, 3, 5, 7}. (i) A ∪ B = {1, 2, 3, 4, 5, 6, 7, 8} = U. So (A ∪ B)' = ∅. Also A' ∩ B' = {2,4,6,8} ∩ {1,3,5,7} = ∅. Both sides equal ∅ ✓. (ii) A ∩ B = ∅. So (A ∩ B)' = U = {1,2,3,4,5,6,7,8}. A' ∪ B' = {2,4,6,8} ∪ {1,3,5,7} = {1,2,3,4,5,6,7,8} = U. Both sides equal U ✓. This confirms both De Morgan's laws.

Compute each side independently and compare. Here A and B happen to be a partition of U, so the verification is particularly clean.

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Q9. Let N be the universal set of natural numbers and let A and B be its subsets given by A = {x : x ∈ N and x < 10} and B = {x : x ∈ N and x is a multiple of 5}. Find the complements of A and B.

Answer: A = {1, 2, 3, 4, 5, 6, 7, 8, 9} (natural numbers less than 10). B = {5, 10, 15, 20, 25, ...} (multiples of 5 in N). A' = N − A = {x : x ∈ N and x ≥ 10} = {10, 11, 12, 13, ...} — every natural number from 10 onwards. B' = N − B = {x : x ∈ N and x is NOT a multiple of 5} = {1, 2, 3, 4, 6, 7, 8, 9, 11, 12, 13, 14, 16, 17, ...} — every natural number that does not end in 0 or 5. Note: complements are taken with respect to the universal set N. Both complements are infinite sets.

Take A' = U – A directly. For the multiples-of-5 case, list a few elements to make the pattern clear.

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Q10. Let U = {2, 4, 6, 8, 10, 12, 14, 16}, A = {2, 4, 6, 8}, B = {10, 12, 14, 16}. Verify (i) (A')' = A, (ii) A ∪ A' = U, (iii) (B')' = B.

Answer: A' = U − A = {10, 12, 14, 16} = B (coincidentally). B' = U − B = {2, 4, 6, 8} = A. (i) (A')' = (B)' = A. Indeed (A')' = A ✓. (ii) A ∪ A' = {2, 4, 6, 8} ∪ {10, 12, 14, 16} = {2, 4, 6, 8, 10, 12, 14, 16} = U ✓. (iii) (B')' = (A)' = B. Indeed (B')' = B ✓. General conclusion: the complement is an INVOLUTION — applying it twice returns the original set. And every element of U lies in exactly one of A or A', confirming A ∪ A' = U and A ∩ A' = ∅.

Compute A', B', then take complements again and verify equality with the original sets.

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